Week 4 Update

Michael Fang

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Topics this week:

- Why α can't exceed 3 (rank-dependent bound)
- What controls the $\bar{\Lambda}$ ranking
- URL sweep, g_2-invariant patterns, metallic chain figures
- Yuan-Torquato improved spreadability for BT
- 138 $|\det M|$=2 chains

Recap

Last week: proved $2<\alpha<3$ gap for $|\det M|=1$ matrices, found first $|\det M|=2$ example at $\alpha = 2.071$, Silver $\bar{\Lambda} = 1/4$, metallic-mean sequence approaching $\sim 1/3$.

Since last meeting:

- Worked out why α can't exceed 3 for $|\det M|=1$ substitutions
- Understood what determines the $\bar{\Lambda}$ ranking
- Swept URL $\bar{\Lambda}(a)$ across $a \in [0, 2]$
- Looked into g_2-invariant patterns from Torquato & Stillinger (2003)
- Generated metallic-mean tile visualizations for $n = 1$ through 10
- Extended metallic-mean $\bar{\Lambda}$ to $n = 100$
- Implemented Yuan-Torquato (2026) improved spreadability fitting
- Computed $\bar{\Lambda}$ for all 138 $|\det M|=2$ chains with $\alpha \in (2, 3)$

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Week 3 catalog for reference:

Pattern	α	$\bar{\Lambda}$
Integer Lattice	—	1/6
Fibonacci	3	0.201
Silver	3	1/4
Bronze	3	0.282
URL (a=1)	2	1/3
BT	1.545	0.377

Why α can't exceed 3

For a rank-r substitution matrix with $|\det M| = 1$ and all $|\lambda_i| < 1$ for $i \geq 2$:

$$\underbrace{\lambda_1 \prod_{i=2}^{r} |\lambda_i|}_{=|\det M|=1} \;\leq\; \lambda_1 \, |\lambda_2|^{r-1} \quad \Rightarrow \quad |\lambda_2| \;\geq\; \lambda_1^{-1/(r-1)}$$

Substituting into $\alpha = 1 - 2\ln|\lambda_2| / \ln \lambda_1$:

$$\boxed{\alpha \;\leq\; 1 + \frac{2}{r-1}}$$

Rank r	$\alpha_{\max}$	Example
2	**3**	all metallic means (Fibonacci, Silver, ...)
3	2	504 cubic matrices with complex pair
4	5/3	
$r \to \infty$	$\to 1^+$	

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The bound is tight for rank 2: metallic means saturate it. For rank 3: saturated by the complex-pair case ($\alpha = 2$ exactly).
Higher rank $\to$ more eigenvalues sharing the determinant constraint $\to$ $|\lambda_2|$ forced larger $\to$ α forced smaller.
For $|\det M| > 1$, the bound generalizes to $\alpha \leq 1 + \frac{2}{r-1}\left(1 - \frac{\ln|\det M|}{\ln \lambda_1}\right)$.

What determines the $\bar{\Lambda}$ ranking?

$\bar{\Lambda}$ measures residual large-scale density fluctuations.
More disorder $\rightarrow$ larger $\bar{\Lambda}$.

- **Lattice** (perfect order): $\bar{\Lambda} = 1/6$, global minimum
- **Metallic means** (tile disparity): as n grows, $L/S = \mu_n$ increases, short tiles become rare defects
  $\Rightarrow \bar{\Lambda}$ increases: $0.201 \rightarrow 0.250 \rightarrow \cdots \rightarrow \sim 1/3$
- **URL** (displacement amplitude): $\bar{\Lambda} = (1 + a^2)/6$, exact. Larger a = more displacement = more disorder
- **Stealthy** (constraint strength): $\bar{\Lambda} \approx 1/(\pi^2 \chi)$. Smaller χ = fewer constrained modes
- **BT** (slow $S(k)$ suppression): $\alpha = 1.545$, close to Class II; large $|\lambda_2| = 0.802 \rightarrow \bar{\Lambda} = 0.377$

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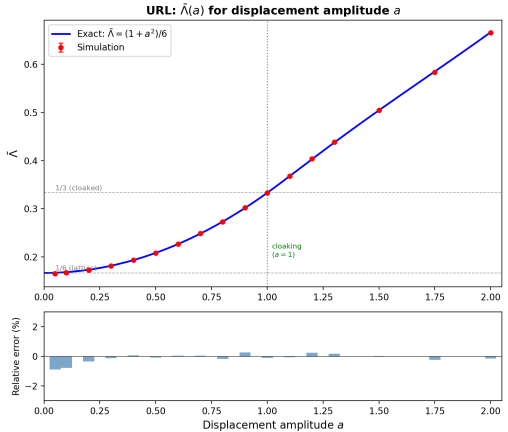
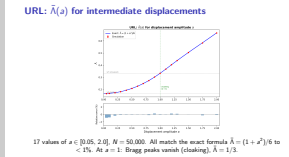
All our patterns fit a single story: more structural disorder = larger $\bar{\Lambda}$.
The metallic-mean sequence is the clearest demonstration: a continuous family with $\bar{\Lambda}$ increasing from 0.201 (Fibonacci) toward $\sim 1/3$.

URL: $\bar{\Lambda}(a)$ for intermediate displacements



17 values of $a \in [0.05, 2.0]$, $N = 50{,}000$. All match the exact formula $\bar{\Lambda} = (1 + a^2)/6$ to $< 1\%$. At $a = 1$: Bragg peaks vanish (cloaking), $\bar{\Lambda} = 1/3$.



Cloaking: the displacement form factor $\tilde{f}(k) = \sin(\pi ka)/(\pi ka)$ vanishes at all non-zero reciprocal lattice vectors when a is an integer.
At $a = 1$: $\bar{\Lambda} = 1/3$, close to where the metallic-mean sequence appears to be heading.
For $a > 1$, displacement intervals overlap. $\bar{\Lambda}$ continues to increase: $\bar{\Lambda}(2) \approx 0.667$ at the second cloaking event.

Klatt et al. (2020) derived the formula analytically.

g_2-invariant patterns

From Torquato & Stillinger (2003): a g_2-invariant process has a fixed pair correlation $g_2(r)$ across a range of densities. At terminal density: $S(0) = 0$ (hyperuniform).

Pattern	Construction	$\bar{\Lambda}$
Step-function g_2	hard rods at close packing	1/6 (lattice)
Lattice-gas	one point uniform per cell	1/3 (= URL a=1)
Gauss. shuffled, σ=0.1	$\mathcal{N}(0, 0.01)$ per site	0.20 ± 0.02
Gauss. shuffled, σ=0.2	$\mathcal{N}(0, 0.04)$ per site	0.250 ± 0.007
Gauss. shuffled, σ=0.3	$\mathcal{N}(0, 0.09)$ per site	0.342 ± 0.001

The lattice-gas is just URL a=1 with a phase offset. Gaussian shuffled lattice at σ=0.1 matches Fibonacci; σ=0.2 matches Silver (1/4).

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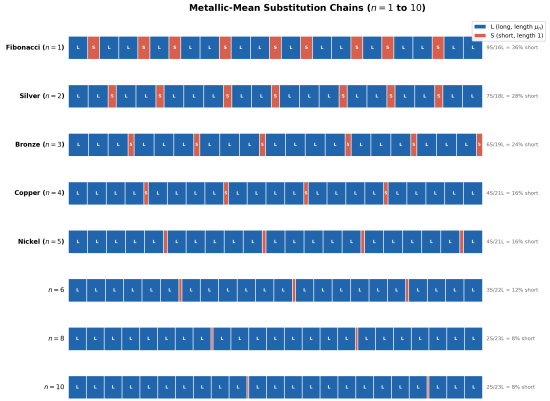
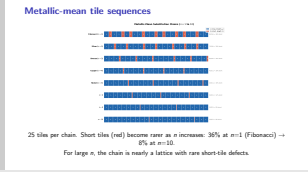
The 1D g_2-invariant processes all reduce to perturbed lattices. "Lattice-gas": tessellate line into unit cells, place one point uniformly in each. This is $j + U(0, 1)$ vs. URL's $j + U(-1/2, 1/2)$. The Gaussian results are interesting: deterministic quasicrystals (Fibonacci, Silver) happen to match specific Gaussian displacements. The structures are completely different, but the large-scale fluctuations are the same.

Metallic-mean tile sequences



25 tiles per chain. Short tiles (red) become rarer as n increases: 36% at n=1 (Fibonacci) → 8% at n=10.

For large n, the chain is nearly a lattice with rare short-tile defects.



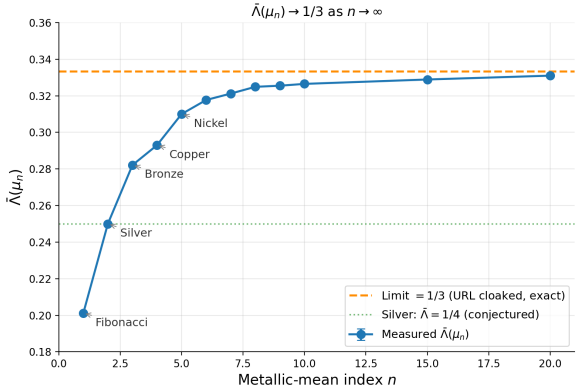
Rules: $S \to L$, $L \to L^n S$.

$n = 1$ (Fibonacci): $S \to L$, $L \to LS$. Balanced quasi-periodic mix.

$n = 10$: $L \to LLLLLLLLLS$. Almost all L tiles, rare S defects.

S fraction $= 1/\mu_n \to 0$ as $n \to \infty$. The chain approaches a lattice with rare quasi-random defects. If the limit is 1/3, this would match URL $a = 1$.

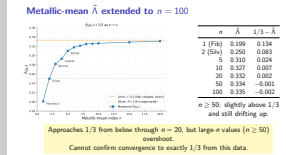
Metallic-mean $\bar{\Lambda}$ extended to $n = 100$



n	$\bar{\Lambda}$	$1/3 - \bar{\Lambda}$
1 (Fib)	0.199	0.134
2 (Silv)	0.250	0.083
5	0.310	0.024
10	0.327	0.007
20	0.332	0.002
50	0.334	−0.001
100	0.335	−0.002

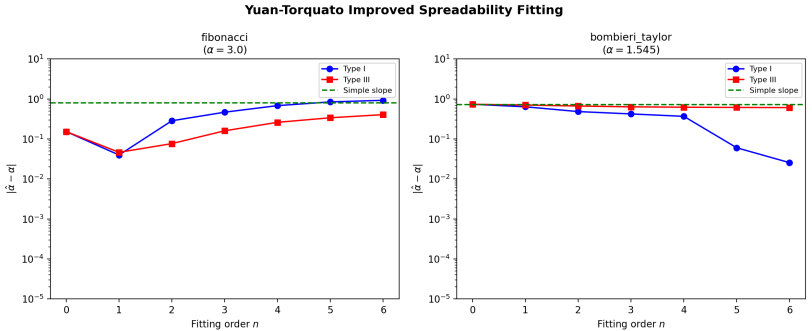
$n \geq 50$: slightly above 1/3 and still drifting up.

Approaches 1/3 from below through $n = 20$, but large-n values ($n \geq 50$) overshoot.
Cannot confirm convergence to exactly 1/3 from this data.



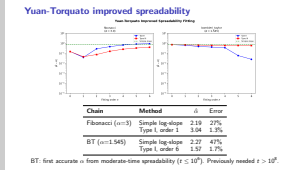
Extended from $n = 20$ (last week) to $n = 100$.
$n = 1 \ldots 20$: monotone increasing toward 1/3 from below. $n = 50$: 0.334, $n = 100$: 0.335 — both slightly above $1/3 = 0.333$.
Three possibilities: 1. True limit is 1/3, overshoots are noise (error bars ~ 0.005, but may be underestimated at large n where only 2 substitution iterations fit in $N = 300$k). 2. True limit is slightly above 1/3. 3. Doesn't converge (unlikely given the smooth trend for $n \leq 20$).
The $n = 1 \ldots 20$ data strongly suggests a limit near 1/3, but higher-precision measurements at large n are needed to determine whether it's exactly 1/3.
Figure shows the original $n = 1 \ldots 20$ data only.

Yuan-Torquato improved spreadability



Chain	Method	$\hat{\alpha}$	Error
Fibonacci (α=3)	Simple log-slope	2.19	27%
	Type I, order 1	3.04	1.3%
BT (α=1.545)	Simple log-slope	2.27	47%
	Type I, order 6	1.57	1.7%

BT: first accurate α from moderate-time spreadability ($t \leq 10^6$). Previously needed $t > 10^8$.



Yuan & Torquato (2026), arXiv:2602.17873.
They add higher-order correction terms to the long-time asymptotic expansion of $s^{ex}(t)$.
Type I: $\ln[s^{ex}] = -(d+\alpha)/2 \cdot \ln t + \ln(\sum C_{i/2}\, t^{-i/2})$. Fitting order n = number of correction terms.
For BT: correction terms from large $|\lambda_2| = 0.802$ are big, so higher-order fits help a lot.
For Fibonacci: $\alpha = 3$ (integer), dense Bragg peaks cause the expansion to diverge at order > 1.
Their Section VI notes that quasicrystals with dense Bragg peaks are not covered by the regular expansion theory. The method is designed for disordered media.

All $|\det M|$=2 chains with $\alpha \in (2,3)$

Searched all 2×2 matrices with $|\det M| = 2$, $|\lambda_2| < 1$, row sums ≤ 10:

α	Chains	$\bar{\Lambda}$ range
2.071	12	[0.32, 2.01]
2.087	10	[0.34, 1.57]
2.175	18	[0.32, 3.36]
2.271	16	[0.34, 3.76]
2.301	18	[0.33, 4.05]
2.362	12	[0.34, 4.80]
2.393	4	[0.34, 5.08]

138 chains total
12 distinct α values
(7 shown)

$\alpha \in [2.07, 2.39]$

$\bar{\Lambda}$ varies by factors of
5–15 *within each α group*

α alone does **not**
determine $\bar{\Lambda}$.

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Expanded from original 28 matrices (row sums ≤ 6) to 138 (row sums ≤ 10).
Formula: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. With $|\det M| = 2$ and $\lambda_1 > 4$: $\alpha \in (2.0, 2.4)$.
Key finding: two matrices with identical eigenvalues can have $\bar{\Lambda}$ differing by a factor of 15. This means α captures only the leading spectral behavior, while $\bar{\Lambda}$ is sensitive to the full spatial arrangement of tiles.
Smallest $\bar{\Lambda} \approx 0.32$–$0.34$ across all groups, close to $1/3$ (URL cloaked value).

Summary

Since last meeting:

- $\alpha \leq 1 + 2/(r-1)$ for rank-r matrices with $|\det M|=1$; $\alpha=3$ is the global maximum (rank 2 only)
- $\bar{\Lambda}$ ranking: controlled by tile disparity, displacement amplitude, spectral constraint strength
- URL sweep confirms $(1 + a^2)/6$ for $a \in [0, 2]$; cloaking at $a=1$
- g_2-invariant patterns: lattice-gas = URL $a=1$; Gaussian shuffled lattice matches Fibonacci/Silver at specific σ
- Metallic-mean tiles visualized ($n=1 \ldots 10$); extended to $n=100$ (approaches $\sim 1/3$ but overshoots)
- Yuan-Torquato: BT α error reduced from 47% to 1.7%
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Possible next steps:

- Write JP results sections (§4–§8)
- Prove Silver $\bar{\Lambda} = 1/4$ via theta-function method
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References

- Torquato & Stillinger (2003), Phys. Rev. E **68**, 041113
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